Matching Solution Method for Joint Eigenfunctions of Cosmological Perturbation Bases

Will Handley   Wei Ning Deng

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Abstract

We address the problem of finding coefficient vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^{N_k}$ such that the time evolutions $X_1^p\boldsymbol{a}$ and $X_2^p\boldsymbol{b}$ are *equal* for every perturbation type p simultaneously, not merely correlated. We show that canonical correlation analysis (CCA, see `cca_formulation.pdf`) is insufficient for this goal because it maximises a single aggregate correlation, allowing strong agreement in some fields to mask a poor match in others. The correct formulation is to minimise the sum of squared per-field residuals, which reduces to the minimum-eigenvalue problem on a $(2N_k \times 2N_k)$ block matrix. The framework is equivalent to finding a vector in the intersection of null spaces, a classical construction in linear algebra.

1 Problem Statement

The setup is the same as in `cca_formulation.pdf`. A Boltzmann code produces solutions for perturbation variables $\delta_r, \delta_m, v_r, v_m$ as functions of conformal time τ and wavenumber k. For N_k wavenumbers we have two bases:

- **Basis 1** (integer-k): $X_1^p \in \mathbb{R}^{N_\tau \times N_k}$,
- **Basis 2** (allowed-k): $X_2^p \in \mathbb{R}^{N_\tau \times N_k}$,

for each perturbation type p.

Goal. Find sparse coefficient vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^{N_k}$ such that

$$X_1^p\boldsymbol{a} = X_2^p\boldsymbol{b} \qquad \text{for all } p. \tag{1}$$

This is a stricter requirement than the CCA goal of maximising the correlation between $X_1^p\boldsymbol{a}$ and $X_2^p\boldsymbol{b}$: we want the two time-series vectors to be *identical*, not merely proportional.

2 Limitation of CCA for Exact Matching

The CCA objective is

$$\rho(\boldsymbol{a}, \boldsymbol{b}) = \frac{\boldsymbol{a}^\top G_{12}\, \boldsymbol{b}}{\sqrt{\boldsymbol{a}^\top G_1\, \boldsymbol{a} \, \cdot \, \boldsymbol{b}^\top G_2\, \boldsymbol{b}}}, \tag{2}$$

where $G_{12} = \sum_p w_p(\bar{X}_1^p)^\top \bar{X}_2^p$ aggregates the cross-field agreement into a *single scalar* via a weighted sum over p.

Two structural problems prevent CCA from guaranteeing exact per-field matching.

Problem 1: outvoting. A high value $\rho \approx 1$ only means that the *weighted average* agreement across all fields is high. If δ_r, v_r, v_m match well but δ_m does not, the three well-matching fields can dominate the sum and push ρ close to 1 while δ_m remains poorly matched. This is precisely what is observed: modes with $\rho = 0.999$ show a systematic offset in δ_m not present in the other three fields.

Problem 2: correlation vs. equality. Even if the per-field contribution of δ_m to ρ were 1, this would only guarantee that $\bar{X}_1^{\delta_m}\boldsymbol{a}$ and $\bar{X}_2^{\delta_m}\boldsymbol{b}$ are *proportional*, i.e. $\bar{X}_1^{\delta_m}\boldsymbol{a} = \lambda\,\bar{X}_2^{\delta_m}\boldsymbol{b}$ for some scalar λ. The Frobenius pre-normalisation further obscures the relationship between the unnormalised time series. Correlation is scale-invariant; equality is not.

3 The Matching Solution Formulation

3.1 Residual objective

Define the per-field residual vector for coefficient pair $(\boldsymbol{a}, \boldsymbol{b})$:

$$\boldsymbol{r}^p(\boldsymbol{a}, \boldsymbol{b}) = \bar{X}_1^p\boldsymbol{a} - \bar{X}_2^p\boldsymbol{b}. \tag{3}$$

The goal (**??**) (after Frobenius pre-normalisation) is $\boldsymbol{r}^p = \boldsymbol{0}$ for all p. We seek non-trivial $(\boldsymbol{a}, \boldsymbol{b})$ that minimise the total squared residual:

$$J(\boldsymbol{a}, \boldsymbol{b}) = \sum_p w_p\|\boldsymbol{r}^p(\boldsymbol{a}, \boldsymbol{b})\|^2 = \sum_p w_p\|\bar{X}_1^p\boldsymbol{a} - \bar{X}_2^p\boldsymbol{b}\|^2. \tag{4}$$

Unlike the CCA correlation, $J = 0$ guarantees (**??**) exactly. The value of J provides an absolute, per-field measure of disagreement: it is not normalised by any self-energy, so a small J cannot be achieved by making $\boldsymbol{a}$ or $\boldsymbol{b}$ small unless they are truly zero.

3.2 Block matrix construction

Expand the squared norm in (**??**):

$$\|\bar{X}_1^p\boldsymbol{a} - \bar{X}_2^p\boldsymbol{b}\|^2 = \boldsymbol{a}^\top G_1^p\boldsymbol{a} - 2\,\boldsymbol{a}^\top G_{12}^p\boldsymbol{b} + \boldsymbol{b}^\top G_2^p\boldsymbol{b}, \tag{5}$$

where the *per-field* Gram matrices are

$$G_1^p = (\bar{X}_1^p)^\top\bar{X}_1^p, \quad G_2^p = (\bar{X}_2^p)^\top\bar{X}_2^p, \quad G_{12}^p = (\bar{X}_1^p)^\top\bar{X}_2^p, \tag{6}$$

each of shape (N_k, N_k). Note these are the same objects used in `cca_formulation.pdf`, but here kept separate per field rather than summed.

Concatenate $\boldsymbol{c} = (\boldsymbol{a}^\top, \boldsymbol{b}^\top)^\top \in \mathbb{R}^{2N_k}$. The objective becomes a quadratic form:

$$J(\boldsymbol{c}) = \boldsymbol{c}^\top M\,\boldsymbol{c}, \tag{7}$$

where $M \in \mathbb{R}^{2N_k\times 2N_k}$ is the symmetric positive semi-definite block matrix

$$M = \sum_p w_p \begin{pmatrix} G_1^p & -G_{12}^p \\ -(G_{12}^p)^\top & G_2^p \end{pmatrix}. \tag{8}$$

Note that M is the weighted sum of matrices of the form $(Z^p)^\top Z^p \succeq 0$ where $Z^p = [\bar{X}_1^p,\, -\bar{X}_2^p]$, so $M \succeq 0$ always. The diagonal blocks of M recover the aggregate Gram matrices $G_1 = \sum_p w_p G_1^p$ and $G_2 = \sum_p w_p G_2^p$ from `cca_formulation.pdf`; the off-diagonal block is $-G_{12} = -\sum_p w_p G_{12}^p$.

3.3 Eigenvalue problem

Minimising $\boldsymbol{c}^\top M\,\boldsymbol{c}$ subject to $\|\boldsymbol{c}\| = 1$ is the standard minimum eigenvalue problem. Since $M \succeq 0$, its eigenvalues $\lambda_1 \le \lambda_2 \le \cdots$ are non-negative, and the minimisers are the eigenvectors corresponding to the smallest eigenvalues:

$$M\,\boldsymbol{c}_i = \lambda_i\,\boldsymbol{c}_i. \tag{9}$$

Each eigenvector $\boldsymbol{c}_i = (\boldsymbol{a}_i^\top, \boldsymbol{b}_i^\top)^\top$ splits into the basis-1 coefficient $\boldsymbol{a}_i = \boldsymbol{c}_i[: N_k]$ and the basis-2 coefficient $\boldsymbol{b}_i = \boldsymbol{c}_i[N_k :]$.

The eigenvalue λ_i is the minimum achievable value of the objective:

$$\lambda_i = J(\boldsymbol{c}_i) = \sum_p w_p \|\bar{X}_1^p \boldsymbol{a}_i - \bar{X}_2^p \boldsymbol{b}_i\|^2. \tag{10}$$

A mode with $\lambda_i \approx 0$ is a valid joint solution for *all* perturbation types simultaneously. Modes with λ_i large are incompatible between the two bases.

To set a threshold, it is convenient to normalise by the maximum eigenvalue and keep modes with

$$\frac{\lambda_i}{\lambda_{\max}} < \lambda_{\min}, \tag{11}$$

for a chosen $\lambda_{\min}$ (e.g. 10^{-3}). Alternatively, the raw value of λ_i can be inspected directly: since the data are Frobenius-normalised, $\lambda_i = 0$ is exact agreement.

3.4 Null space interpretation

The condition (**??**) is equivalent to

$$Z^p\,\boldsymbol{c} = \boldsymbol{0} \qquad \text{for all } p, \qquad Z^p = [\bar{X}_1^p,\, -\bar{X}_2^p] \in \mathbb{R}^{N_\tau \times 2N_k}. \tag{12}$$

So the exact solutions lie in the intersection of null spaces:

$$\boldsymbol{c} \in \ker(Z^{\delta_r}) \cap \ker(Z^{\delta_m}) \cap \ker(Z^{v_r}) \cap \ker(Z^{v_m}). \tag{13}$$

In general this intersection may be trivial (containing only $\boldsymbol{0}$), in which case no exact joint eigenfunction exists. The minimum-eigenvalue problem finds the vector $\boldsymbol{c}$ that is "closest" to being in all null spaces simultaneously, in the least-squares sense.

4 Relationship to CCA

The CCA and the matching-solution method optimise different objectives:

	CCA	Matching solution
Objective	Maximise $\rho(\boldsymbol{a}, \boldsymbol{b}) \in [0, 1]$ (correlation)	Minimise $J(\boldsymbol{a}, \boldsymbol{b}) \geq 0$ (squared difference)
Found when	$\boldsymbol{a} \propto \boldsymbol{b}$ (same shape)	$X_1^p \boldsymbol{a} = X_2^p \boldsymbol{b}$ (same value)
Quality metric	$\rho \to 1$ (high = good)	$\lambda \to 0$ (low = good)
Outvoting	Yes: 3 fields can hide δ_m mismatch	No: J penalises every field independently
Scale sensitivity	No (correlation is scale-invariant)	Yes ($\lambda = 0$ requires exact equality)
Matrix size	$(N_k \times N_k)$ Gram matrices	$(2N_k \times 2N_k)$ block matrix

Theorem 1. *If $X_1^p \boldsymbol{c}_0 = X_2^p \boldsymbol{c}_0$ for all p for some $\boldsymbol{c}_0 \neq \boldsymbol{0}$, then $\boldsymbol{c} = (\boldsymbol{c}_0^\top, \boldsymbol{c}_0^\top)^\top / \|\boldsymbol{c}_0\|$ achieves $\lambda = 0$ in the matching-solution method and $\rho = 1$ in CCA (when $\|X_1^p\|_{\mathrm{F}} = \|X_2^p\|_{\mathrm{F}}$ for all p).*

Proof. Direct substitution: $J(\boldsymbol{c}) = \sum_p w_p \|\bar{X}_1^p \boldsymbol{c}_0 / \|\boldsymbol{c}_0\| - \bar{X}_2^p \boldsymbol{c}_0 / \|\boldsymbol{c}_0\|\|^2 = 0$, so $\lambda = 0$. The CCA claim follows from Theorem 1 of `cca_formulation.pdf`. $\square$

The converse does not hold for CCA: $\rho = 1$ implies only proportionality, not equality. The converse *does* hold for the matching-solution method: $\lambda = 0$ and $\boldsymbol{c} \neq \boldsymbol{0}$ imply $\bar{X}_1^p \boldsymbol{a} = \bar{X}_2^p \boldsymbol{b}$ for all p.

5 Sparse Basis via Oblique Rotation

The eigenvectors $\boldsymbol{c}_i$ are dense in general. Let $A = [\boldsymbol{a}_1 \mid \cdots \mid \boldsymbol{a}_m]$ and $B = [\boldsymbol{b}_1 \mid \cdots \mid \boldsymbol{b}_m]$ collect the m valid coefficient vectors. We seek an invertible $M \in \mathbb{R}^{m \times m}$ such that the rotated columns of $\tilde{A} = AM^\top$ are sparse (concentrated on one or a few k-modes), applying the same rotation to B.

The same analysis as in `cca_formulation.pdf` applies: Varimax is inappropriate (the $\boldsymbol{a}_i$ are not orthonormal in k-space), while Promax oblique rotation or FastICA are recommended.

6 Complete Pipeline

Algorithm 1 Matching-solution sparse joint eigenfunction method

Input: $X_1^p, X_2^p \in \mathbb{R}^{N_\tau \times N_k}$ for each p; weights $\{w_p\}$; threshold $\lambda_{\min}$

Frobenius normalise

  $\bar{X}_\alpha^p \leftarrow X_\alpha^p / \|X_\alpha^p\|_{\mathrm{F}}$ for all α, p

Compute per-field Gram matrices ▷ Eq. (??)

  $G_1^p \leftarrow (\bar{X}_1^p)^\top \bar{X}_1^p$ for each p

  $G_2^p \leftarrow (\bar{X}_2^p)^\top \bar{X}_2^p$

  $G_{12}^p \leftarrow (\bar{X}_1^p)^\top \bar{X}_2^p$

Construct block matrix ▷ Eq. (??)

  $M \leftarrow \sum_p w_p \left[\begin{smallmatrix} G_1^p & -G_{12}^p \\ -(G_{12}^p)^\top & G_2^p \end{smallmatrix}\right]$ ▷ $(2N_k, 2N_k)$

Solve eigenvalue problem ▷ Eq. (??)

  $M\,\boldsymbol{c}_i = \lambda_i\,\boldsymbol{c}_i, \quad \lambda_1 \le \lambda_2 \le \cdots$

  Keep m eigenvectors with $\lambda_i/\lambda_{\max} < \lambda_{\min}$

  $\boldsymbol{a}_i \leftarrow \boldsymbol{c}_i[:N_k], \quad \boldsymbol{b}_i \leftarrow \boldsymbol{c}_i[N_k:]$ for $i = 1, \ldots, m$

Oblique rotation for sparsity ▷ §??

  $A \leftarrow [\boldsymbol{a}_1 \mid \cdots \mid \boldsymbol{a}_m], \quad B \leftarrow [\boldsymbol{b}_1 \mid \cdots \mid \boldsymbol{b}_m]$

  $\tilde{A}, \tilde{B} \leftarrow \mathrm{Promax}(A, B)$ ▷ or FastICA

Output: $\tilde{A}, \tilde{B} \in \mathbb{R}^{N_k \times m}$

  Each column pair $(\tilde{\boldsymbol{a}}_j, \tilde{\boldsymbol{b}}_j)$: sparse coefficients satisfying $X_1^p \tilde{\boldsymbol{a}}_j \approx X_2^p \tilde{\boldsymbol{b}}_j$ for all p

7 Discussion

1. **Stricter criterion.** The eigenvalue λ_i measures the total squared difference across all perturbation types. No field can be "outvoted": even a small mismatch in a single field contributes positively to λ_i. This directly addresses the δ_m mismatch observed with the CCA method.

2. **Larger matrix.** The block matrix M is $(2N_k \times 2N_k)$ compared to $(N_k \times N_k)$ for the CCA Gram matrices. For $N_k \lesssim 10^3$ this remains computationally cheap; only the smallest few eigenvalues need to be computed (e.g. via `scipy.sparse.linalg.eigsh`).

3. **Separate coefficient vectors.** The method naturally produces $\boldsymbol{a}_i \neq \boldsymbol{b}_i$ in general. For exact joint eigenfunctions the two coincide; for approximate ones (λ_i small but non-zero) they may differ slightly, and both are meaningful.

4. **Weighting.** The weights w_p encode the relative importance of each perturbation type. Unlike in CCA, a larger weight for δ_m now directly penalises δ_m mismatch rather than merely up-weighting its contribution to an aggregate correlation. Setting all $w_p = 1$ gives equal treatment to all fields.

5. **Relation to CCA.** The CCA method can be recovered approximately from the matching-solution method. When the two bases are similar ($\|X_1^p\|_{\mathrm{F}} \approx \|X_2^p\|_{\mathrm{F}}$), modes with $\lambda_i \approx 0$ also achieve $\rho_i \approx 1$, but the converse does not hold. Use CCA when shape agreement is sufficient; use the matching-solution method when exact pointwise equality of time evolutions is required.